EASTERN INTERNATIONAL UNIVERSITY

BECAMEX BUSINESS SCHOOL



"Exploratory Data Analysis: Supermarket Sales Performance "

COURSE: MIS 311 - Introduction to Business Analytics

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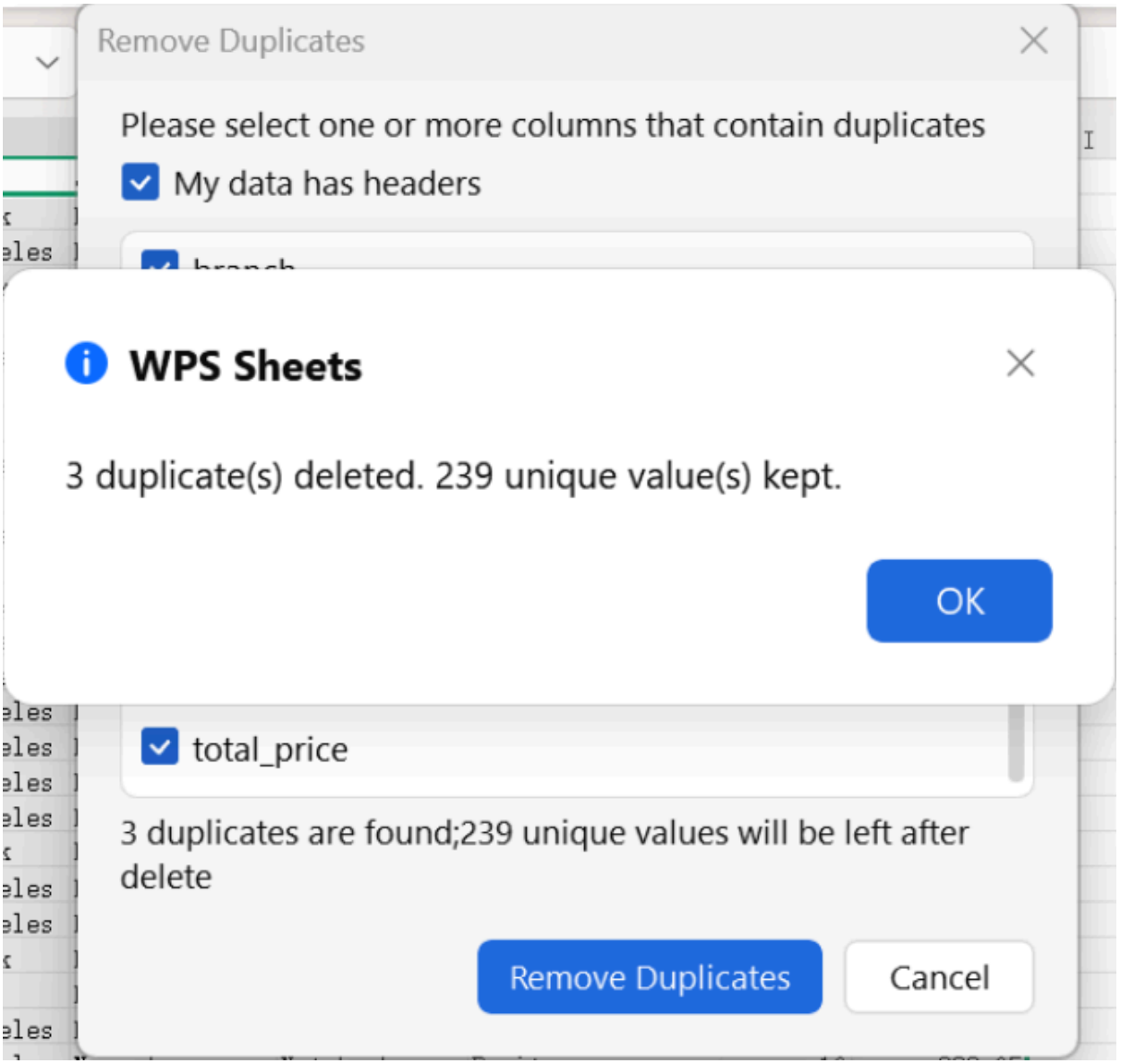
Part 1: Data Analysis and Insight Generation

1. Data Overview

- **Context:** This report analyzes the "10_Supermarket Sale" dataset, which records transaction details from a supermarket chain, including branch locations, customer types, product categories, quantities sold, and total prices. This data is relevant for understanding retail performance and customer purchasing behavior.
- **Dataset Size:** The original dataset consists of 253 rows and 8 columns.

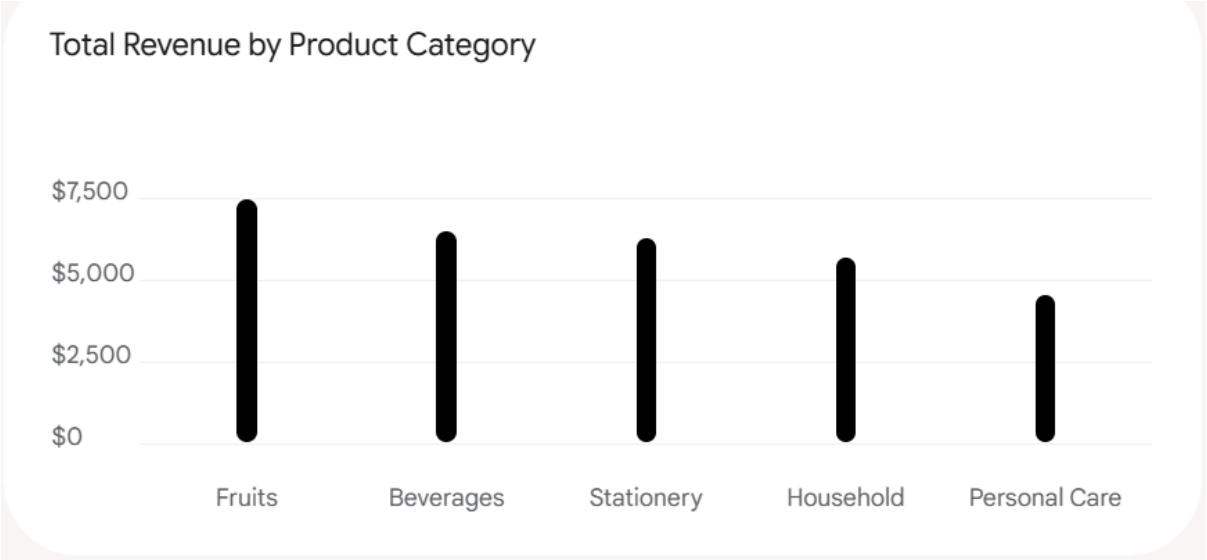
2. Data Cleaning To ensure the accuracy of the analysis, the following data cleaning steps were performed:

- **Handling Duplicates:** 3 duplicate rows were identified and removed.
- **Handling Missing Values:** Missing values were found in the customer_type, product_category, and quantity columns. Because these incomplete records represent a very small portion of the dataset, they were removed to maintain data integrity.
- **Final Dataset:** After cleaning, the final dataset contains 239 valid observations ready for analysis.



3. Descriptive Statistics Descriptive statistics were calculated for the numerical variables across the 239 valid transactions:

- The average (mean) quantity purchased per transaction is 10.78 items, ranging from a minimum of 1 item to a maximum of 20 items.
- The average transaction revenue (mean total price) is $127.04.
- The transaction values show high variance, with the lowest bill being $2.18 and the highest bill reaching $427.14 (Standard Deviation = 102.80).



4. Key Insights Based on the descriptive statistics and data visualization, two key business insights are derived:

- **Insight 1: Product Category Performance & Inventory Strategy** The analysis reveals that Fruits ($7,450.12) and Beverages ($6,463.17) are the top-performing categories, whereas Personal Care ($4,509.59) generates the lowest revenue. **Business Implication:** The supply chain management should prioritize inventory replenishment for Fruits and Beverages to prevent stockouts, while the marketing team should investigate pricing strategies or promotions to stimulate sales for Personal Care products.
- **Insight 2: The Value of the Membership Program** By grouping revenue by customer type, the data shows that registered Member customers contributed significantly more to total sales ($17,614.08) compared to Normal customers ($12,748.86). **Business Implication:** The store's loyalty program is highly effective. Management should invest in targeted campaigns to convert "Normal" customers into "Members," thereby increasing customer retention and long-term profitability.